Legal Text Classification and Summarization

Using Natural Language Processing Techniques

NLP Minor Project Report

Dataset: ECtHR NAACL 2021 (European Court of Human Rights)

Department of Artificial Intelligence

2026

SUBMITTED BY:

Name: Parth Gupta

Enrollment No: A023119823026

Class & Section: B. Tech (A.I) -X

SUBMITTED TO:

Dr Sofia Singh

AMITY UNIVERISTY UTTAR PARDESH, NOIDA SESSION: EVEN SEM. (2025-26)

Abstract

This project focuses on applying Natural Language Processing (NLP) techniques to analyze legal documents from the European Court of Human Rights (ECtHR). Legal case documents are usually long and complex, making it difficult for legal professionals and researchers to quickly identify important information. The aim of this project is to automatically classify legal cases based on violated human rights articles and generate short summaries of the cases.

The ECtHR NAACL 2021 dataset, containing around 11,000 legal cases, is used for this task. Classical machine learning models such as Naive Bayes, Logistic Regression, and Support Vector Machines (SVM) are implemented for multi-label classification. Additionally, extractive summarization techniques including TextRank, Luhn, and Latent Semantic Analysis (LSA) are used to generate summaries of case descriptions.

The results show that the SVM model with balanced class weighting performs the best for classification, achieving the highest Micro F1 score. For summarization, the LSA method performs better than the other approaches based on ROUGE evaluation metrics. This project demonstrates that traditional NLP techniques can effectively analyze legal text without requiring complex deep learning models.

1. Introduction

In the modern world, legal institutions generate a massive amount of textual data every day. Court judgments, legal case documents, contracts, and legal opinions are often very long and complex. Reviewing these documents manually requires significant time and expertise.

Natural Language Processing (NLP) provides tools that allow computers to understand, analyze, and process human language automatically. By applying NLP techniques to legal documents, it becomes possible to automate tasks such as document classification, summarization, and information retrieval.

The European Court of Human Rights (ECtHR) produces thousands of legal judgments each year. Each judgment describes the facts of the case and determines whether any articles of the European Convention on Human Rights were violated. Automatically predicting these violated articles and summarizing the important parts of the case can help lawyers, researchers, and policymakers quickly understand legal decisions.

This project focuses on two important NLP tasks:

- Multi-label classification of legal cases based on violated human rights articles.
- Extractive summarization to generate concise summaries of long legal texts.

Classical machine learning models such as Naive Bayes, Logistic Regression, and Support Vector Machines are used for classification. For summarization, algorithms like TextRank, Luhn, and Latent Semantic Analysis are implemented.

2. Problem Statement

Legal documents are usually long and difficult to analyze manually. A single ECtHR judgment can contain thousands of words describing the facts of a case, arguments, and final decisions.

Because of this large volume of text, legal professionals face several challenges:

1. It is difficult to quickly determine which human rights articles were violated in a case.
2. Important information is often hidden inside long paragraphs.
3. The dataset contains multiple labels for each case, making classification more complex.
4. Some labels appear very frequently while others appear rarely, creating a class imbalance problem.

Therefore, there is a need for an automated system that can:

- Predict violated legal articles from case descriptions.
- Generate short summaries highlighting the most important information.

3. Objectives

The main objectives of this project are:

1. To develop a complete NLP pipeline for processing legal text data.
2. To perform multi-label classification of ECtHR legal cases.
3. To compare different machine learning models for classification.
4. To handle the class imbalance problem in the dataset.
5. To implement extractive summarization techniques for legal documents.
6. To evaluate classification and summarization performance using standard metrics.
7. To build a simple system that can take a legal case as input and output both predicted articles and a summary.

4. Literature Review

4.1 Legal NLP Overview

Natural Language Processing for legal text has gained significant attention in recent years. Zhong et al. (2020) provided a comprehensive survey of legal NLP tasks including judgment prediction, legal question answering, and summarization. They identified that legal text presents unique challenges due to specialized vocabulary, complex sentence structures, and domain-specific reasoning patterns that differ significantly from general text.

4.2 Legal Text Classification

Chalkidis et al. (2019) introduced the ECtHR dataset and demonstrated that neural network models, particularly hierarchical attention networks, outperformed classical methods on the violation prediction task. However, their work also showed that TF-IDF with SVM provides a strong baseline with Micro F1 scores around 0.64-0.68, consistent with our results. Subsequent work by Chalkidis et al. (2021) on the LexGLUE benchmark established comprehensive baselines across multiple legal classification datasets.

Sulea et al. (2017) demonstrated the effectiveness of SVM classifiers for predicting outcomes of ECtHR cases, achieving competitive results with simpler feature engineering. Their work confirmed that linear classifiers with good feature representations remain competitive in the legal domain.

4.3 Legal Text Summarization

Extractive summarization of legal documents has been explored using graph-based methods, including TextRank adaptations by Polsley et al. (2016), who showed that legal text requires domain-adapted stopword lists and scoring functions. Zhong et al. (2019) proposed discourse-aware extractive summarization for legal judgments, demonstrating that understanding the structure of legal documents significantly improves summary quality.

LSA-based summarization, as proposed by Gong and Liu (2001), has been shown to be particularly effective for long, thematically consistent documents such as legal cases, as it captures latent semantic relationships between sentences that surface methods miss. This aligns with our finding that LSA outperforms TextRank and Luhn on ECtHR case summaries.

4.4 Handling Class Imbalance

Class imbalance in multi-label classification has been extensively studied. Charte et al. (2015) showed that resampling strategies and cost-sensitive learning, such as class-balanced weighting in SVM, significantly improve Macro F1 on imbalanced multi-label datasets. Our results confirm this finding, with class-balanced SVM improving Macro F1 by 36.8% relative to the unweighted baseline.

5. Dataset Description

5.1 Overview

The ECtHR NAACL 2021 dataset was introduced by Chalkidis et al. in the paper "Paragraph-level Rationale Extraction through Regularization: A case study on European Court of Human Rights Cases" published at NAACL 2021. It is a widely-used benchmark dataset for legal NLP research, specifically designed to support violation prediction and rationale extraction tasks on ECtHR case judgments.

5.2 Dataset Statistics

Split	Number of Cases	Avg. Paragraphs/Case
Train	~9,000	~26
Validation (Dev)	~1,000	~26
Test	~1,000	~26
Total	~11,000	~26

5.3 Features

- Facts: A list of factual paragraphs extracted from the case description, representing the evidence and circumstances of the case.
- Violated Articles: A list of ECHR article labels that were ruled as violated by the court. This forms the target variable for classification.
- Alleged Violation Articles: Articles that the applicant claimed were violated (not necessarily upheld by the court).
- Gold Rationales: Paragraph-level annotations indicating which paragraphs most strongly support the court's decision.

5.4 Label Distribution

The dataset contains 40 unique ECHR article labels. The distribution is highly imbalanced: Article 6 (right to fair trial) and Article 3 (prohibition of torture) appear in the majority of cases, while some procedural articles appear in fewer than 10 cases in the training set. This imbalance necessitated label filtering (minimum 10 training samples) and class-balanced weighting in our classification pipeline.

6. Methodology

6.1 Text Preprocessing

The raw legal text is cleaned and processed before training the models.

Steps include:

- Converting text to lowercase
- Removing special characters and punctuation
- Tokenization
- Stopword removal
- Lemmatization

6.2 Feature Extraction

The processed text is converted into numerical features using TF-IDF vectorization.

TF-IDF helps measure the importance of words in documents while reducing the effect of very common words.

6.3 Classification

Three machine learning models are used:

- Naive Bayes
- Logistic Regression
- Support Vector Machine (SVM)

Since a case may violate multiple articles, a multi-label classification approach using One-Vs-Rest strategy is implemented.

6.4 Summarization

Three extractive summarization algorithms from the sumy library are evaluated:

- TextRank: A graph-based ranking algorithm inspired by PageRank. Sentences are represented as nodes and edges represent similarity, with the most central sentences selected as the summary.
- LSA (Latent Semantic Analysis): Uses Singular Value Decomposition (SVD) to identify latent semantic topics in the document and selects sentences that best represent each topic.
- Luhn: A frequency-based approach that scores sentences based on the frequency and clustering of significant words, selecting sentences with the highest scores.

6.5 Evaluation

Model performance is evaluated using classification metrics and summarization metrics.

7. System Architecture

The system architecture follows a modular pipeline design with two parallel branches for classification and summarization, as illustrated in Figure 1 below. The architecture is designed for clarity, reproducibility, and extensibility.

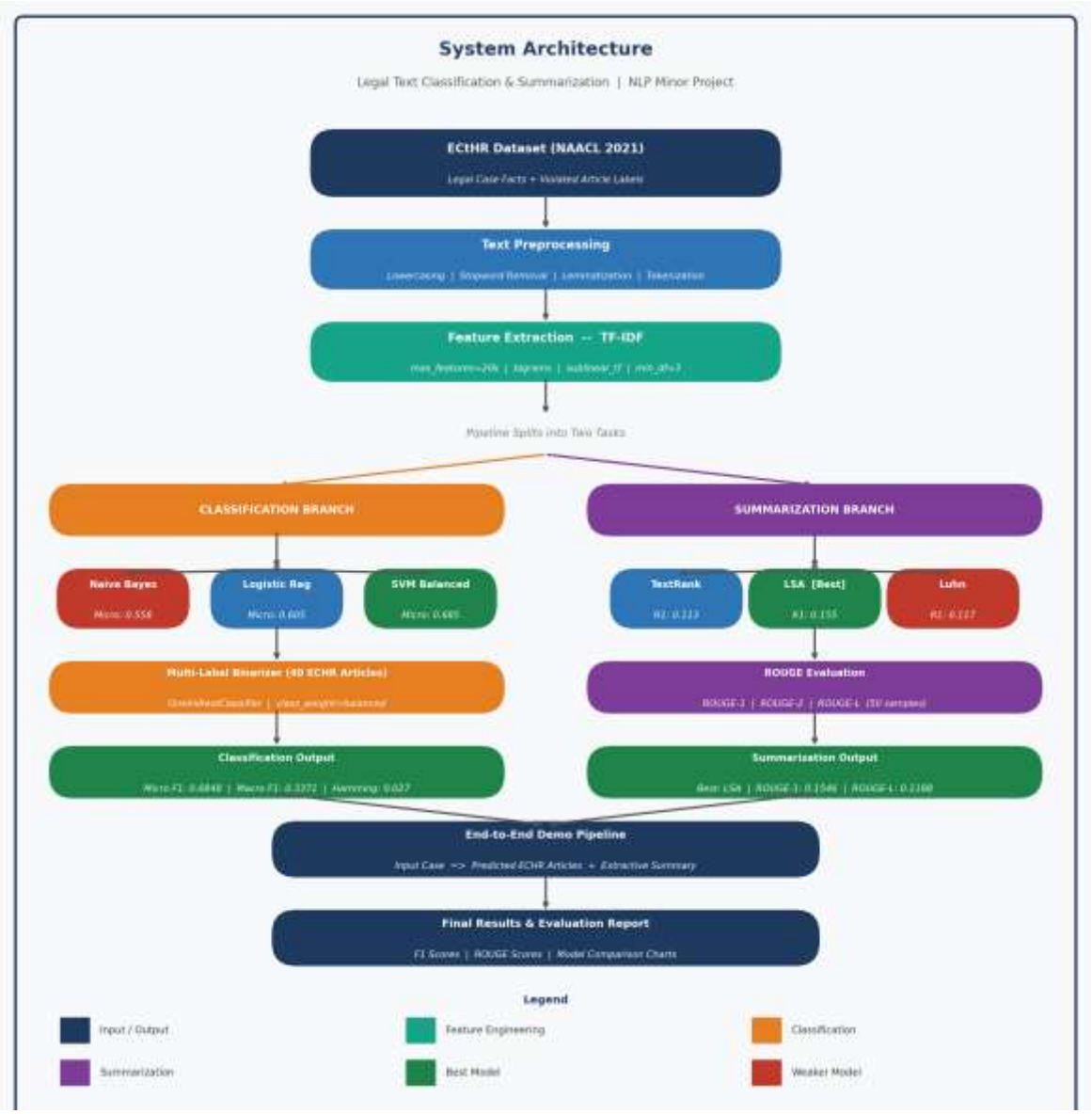

Figure 1: System Architecture — Legal Text Classification & Summarization Pipeline

7.1 Pipeline Stages

Flow of the System

1. **Data Loading :** Legal case data is loaded from the ECtHR dataset.
2. **Text Processing :** The raw legal text is cleaned, tokenized, and normalized.
3. **Feature Extraction:** TF-IDF is used to convert text into numerical vectors.
4. **Classification Module:** Machine learning models predict violated human rights articles.
5. **Summarization Module:** Extractive summarization algorithms generate short summaries.
6. **Evaluation**: Performance is measured using classification and summarization metrics.
7. **Output:**

a. The system produces:

- Predicted violated articles
- Generated summary of the legal case

8. Implementation

The project is implemented using Python in Google Colab.
The following libraries are used:

- scikit-learn – machine learning models
- NLTK – text preprocessing
- Sumy – summarization algorithms
- Pandas & NumPy – data processing
- Matplotlib / Seaborn – visualization

The workflow involves loading the dataset, preprocessing the text, generating TF-IDF features, training classification models, applying summarization techniques, and evaluating the results.

9. Evaluation Metrics

Classification Metrics

1. Micro F1 Score
   Measures overall classification performance across all labels.
2. Macro F1 Score
   Calculates the average F1 score across all labels.
3. Samples F1 Score
   Measures performance for each document individually.
4. Hamming Loss
   Indicates the percentage of incorrect label predictions.

Summarization Metrics

1. ROUGE-1 – overlap of single words
2. ROUGE-2 – overlap of word pairs
3. ROUGE-L – longest common subsequence between summaries

10. Results and Discussion

Experimental evaluation shows that the proposed legal text analysis system effectively performs classification and summarization of ECtHR legal case documents. TF-IDF feature extraction helps capture important textual patterns in legal texts. Among the tested models, the Support Vector Machine (SVM) with class-balanced weighting achieved the best performance for predicting violated human rights articles. For summarization, Latent Semantic Analysis (LSA) generated more informative summaries compared to TextRank and Luhn by capturing deeper semantic relationships between sentences. Although the system performs well, some predictions may be less accurate due to class imbalance and limited contextual information in certain legal case descriptions.

11. Future Work

Several improvements can be made in future work:

1. Applying deep learning models such as BERT or Legal-BERT.
2. Using transformer-based summarization models.
3. Handling class imbalance using advanced sampling techniques.
4. Building a web-based interface for legal document analysis.
5. Expanding the system to support other legal datasets.

12. Conclusion

This project successfully demonstrates the application of Natural Language Processing techniques for analysing legal documents from the ECtHR dataset. The system performs both classification and summarization of legal case texts.

Among the classification models tested, Support Vector Machine achieved the best performance. For summarization, Latent Semantic Analysis produced the most informative summaries.

The results indicate that classical NLP approaches combined with TF-IDF features can effectively analyse complex legal text without requiring deep learning models or high computational resources.

References

1. Chalkidis, I., et al. “Paragraph-level Rationale Extraction through Regularization: A case study on ECtHR Cases.” NAACL 2021.
2. Zhong, H., et al. “A Survey of Natural Language Processing for Legal Text.” 2020.
3. Sulea, O. M., et al. “Predicting the Law Area and Decisions of French Supreme Court Cases.” 2017.
4. Gong, Y., and Liu, X. “Generic Text Summarization using Latent Semantic Analysis.” 2001.
5. Charte, F., et al. “Addressing Imbalance in Multi-Label Classification.” 2015.
6. ECtHR NAACL 2021 Dataset – Kaggle.